Aspect based Sentiment Analysis on Hotel Reviews

UE24CS352A Machine Learning | Team 22 | Project 1
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Problem and scope

Hotel reviews can praise service while criticizing rooms or cleanliness. This project predicts a 1-5 rating for the overall review and each supported hotel aspect, then maps 1-2 to negative, 3 to neutral and 4-5 to positive. The implemented pipeline runs through training, evaluation and an interactive demo. Real-data overall polarity accuracy is **74.6%**; software verification passes, but prediction errors and uneven aspect performance remain.

Dataset details

The primary dataset is the published **ArguAna TripAdvisor annotated v2** corpus [1]: 2,100 English hotel reviews from 122 hotels. Review text and title are taken from the XMI Sofa text; overall labels come from Sentiment.score, not the hotel star category. Aspect ratings come from the Aspects fields. Author metadata is omitted. The download checksum and CSV hash are recorded for reproducibility. Scientific use only; not for commercial applications.

Target	Observed ratings	Target	Observed ratings
Overall	2,100	Cleanliness	1,651
Service	1,608	Rooms	1,647
Location	1,238	Value	1,631
Business service	772	Sleep quality	0

Source -1 labels become missing values and are excluded per target. Sleep quality remains unavailable in real-data mode; no labels are fabricated. The original 320-row template-based sample is retained in a separate synthetic demonstration mode, with all eight targets. Its generating code and external provenance were not supplied; its high scores do not establish real-world performance.

Approach and evaluation design

Reviews are normalized, lowercased and tokenized while preserving negation. Three fixed pipeline configurations are compared per target: TF-IDF plus Multinomial Naive Bayes (alpha 0.5), count unigrams/bigrams plus balanced Linear SVM (C 0.1), and TF-IDF plus balanced Logistic Regression (C 4). Vocabulary is capped at 12,000 features. Because feature representations differ, this is a pipeline comparison rather than a pure classifier ablation.

Seed 42 creates disjoint hotel groups: **1342 training, 388 validation and 370 test reviews**. Nominal 60/20/20 group proportions yield unequal review proportions. Features are fitted on training data only during selection. Validation polarity macro-F1 selects a configuration per target; star accuracy breaks ties. Each selected model is refitted on training plus validation before test evaluation. Saved models never include test reviews.

Implementation

prepare_data.py converts the corpus; src/utils.py validates text and labels; src/train.py writes metrics, splits, confusion matrices and error examples; src/predict.py verifies and loads models. run_project.py offers separate training and inference paths. The Streamlit interface supports both datasets and suppresses unmentioned aspects with a documented keyword heuristic. This heuristic is not a learned aspect extractor and is not applied to model evaluation.

Results and verification

All results below use real reviews from held-out hotels. Missing aspect labels reduce the test count for each target. Values are regenerated from saved predictions.

Target	Model	Test n	Star acc.	Polarity acc.	Macro-F1	Baseline acc.
Overall	NB	370	51.1%	74.6%	0.709	32.4%
Service	LR	271	44.6%	68.3%	0.628	45.0%
Rooms	LR	276	39.1%	60.9%	0.560	47.8%
Location	SVM	223	47.5%	74.4%	0.423	78.9%
Value	NB	279	46.2%	62.4%	0.615	37.3%
Cleanliness	LR	278	46.4%	64.7%	0.539	61.5%
Business Service	LR	130	38.5%	55.4%	0.558	39.2%

Macro-F1 averages negative, neutral and positive F1 equally. The polarity baseline predicts the most frequent development-set polarity; the star baseline separately predicts the most frequent development-set star rating. No baseline learns from test labels.

Interpretation and limitations

Overall polarity accuracy is 74.6%, macro-F1 is 0.709 and exact star accuracy is 51.1%. Aspect polarity accuracy ranges from 55.4% to 74.4%. Six of seven trained targets beat the majority polarity baseline on accuracy. Location scores 74.4% versus a 78.9% majority baseline, with macro-F1 only 0.423: accuracy hides weak minority-class performance. The model should not be described as uniformly superior.

Aspect-specific models outperform transferring the overall prediction for rooms, location and cleanliness, but not service, value or business service. Whole-review features can mix opinions between aspects. Other limits are small labelled samples, noisy user ratings, no real sleep-quality supervision, one fixed split, historical English reviews and an unevaluated keyword mention filter. The corpus is balanced by overall sentiment and is not a representative sample of current hotel traffic. Wilson intervals in the CSV are descriptive per-review intervals; they do not adjust for within-hotel dependence.

Functional and statistical testing

42 automated tests passed; 0 failures and 0 errors. Tests cover invalid text and ratings, duplicate/conflicting rows, reproducible hotel-disjoint partitions, training-only vocabulary, validation-only selection, per-target missing labels, model serialization and hash checks, CLI inference without retraining, unknown vocabulary, and Streamlit prediction/error flows. Final model predictions and both accuracy and macro-F1 are recomputed and checked against the submitted CSV files. The polarity baseline is independently checked against development-set counts.

Run **python verify_project.py** for the saved verification log. Run **python run_project.py** to reproduce training; run **python -m streamlit run streamlit_app.py** for the live demonstration. Python 3.11 and exact tested package versions are documented. Passing functional tests confirms implementation behavior, not perfect sentiment prediction.

Conclusion

The project provides a working, reproducible supervised ABSA prototype with honest held-out evaluation and an offline-capable demo after setup. Future work should add real sleep-quality labels, expand hotel diversity, use development-only cross-validation and evaluate contextual aspect extraction on newly reserved data. The Stanford reference [2] motivates the task; this smaller-corpus experiment is not a reproduction of its reported scores.

References

[1] Wachsmuth et al. (2014). A Review Corpus for Argumentation Analysis. CICLing. Corpus and usage terms: zenodo.org/records/3973241.
[2] Yu, Y. (2016). Aspect-based Sentiment Analysis on Hotel Reviews. Stanford CS229: cs229.stanford.edu/proj2016spr/report/032.pdf.